Machine Learning-Based Phishing Website Detection

Joselyn Carvajal

October 2026

1. Project Title

Machine Learning-Based Phishing Website Detection

2. Kaggle Dataset

Dataset Name: Web Page Phishing Dataset

Source: `https://www.kaggle.com/datasets/danielfernandon/web-page-phishing-dataset`

The dataset contains 100,077 records and 20 columns, including 19 numerical URL features and one phishing classification label.

3. Security Problem

Phishing websites are designed to deceive users into sharing sensitive information, such as passwords and financial details. Detecting these websites is important for preventing credential theft and online fraud.

This project investigates whether machine learning can identify phishing websites based on URL characteristics.

4. Research Question

Can Decision Tree and Random Forest machine learning models accurately distinguish phishing websites from legitimate websites using URL characteristics?

5. Proposed Methodology

The dataset will be analyzed using Python in Google Colab. Exploratory data analysis will examine class distribution, missing values, duplicate records, and relationships between URL features.

Data preprocessing will include removing duplicate records and separating the dataset into training and testing sets.

Two machine learning models, Decision Tree and Random Forest, will be trained and compared.

Their performance will be evaluated using accuracy, precision, recall, F1-score, and confusion matrices.

6. Expected Results

The project is expected to demonstrate that URL characteristics can help machine learning models identify potential phishing websites.

Features such as URL length and the number of slashes may be useful indicators of phishing activity.

The comparison will help determine which model provides more effective phishing detection while considering false positives and false negatives.